

Yash Vardhan Singh

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I build AI-powered products, embedded systems and developer tools, and lead a 1,500+ member student tech community. Currently CTO of Ashoka's Ministry of Academic Affairs, and 2nd place at Startup Ashoka 2026.

Education

Ashoka University | B.Sc. (Liberal Arts), Computer Science & Entrepreneurship | Class of 2030

- Qualified for the Indian Olympiad Qualifier in Mathematics (IOQM)

Experience

Chief Technology Officer, Ministry of Academic Affairs, Ashoka University Student Government | Sept 2026 - Present

- Lead technology for the ministry that represents students' academic interests to the Office of Academic Affairs and shares academic policy and resources with the student body
- Built React websites that serve Ashoka's entire ~4,000-student community from one place
- Developing AI-powered features for the ministry's student-facing tools

Founder, Bits&Bytes (Gobitsnbytes.org)

- Built a 1,500+ member student tech community running projects, workshops and hackathons; it has spread to 10+ Indian cities as "Bits&Bytes forks"
- Built official websites for Gobitsnbytes.org, [Daydream Lucknow](https://DaydreamLucknow.org) and IndiaInnovates.org

Software Engineer Intern, Maximally

- Built MailAF, an AI-powered cold-outreach tool that automates contact research, email drafting and sending (Next.js, Firebase, Prisma, LLMs)

Projects

LISA: Light-based In-field Spectral Analyser | Startup Ashoka 2026, 2nd Place (₹35,000) | github.com/yashclouded/lisa

- Sole developer on a team project: a frugal, Foldscope-style smartphone spectrometer for testing drinking water, aimed at the field kits used under India's Jal Jeevan Mission, which rely on subjective visual colour matching and leave no digital record
- Designed a clip-on origami card enclosure (30×30 mm optical tube, dual razor-blade slit, diffraction grating film) that costs about ₹1,500 per unit
- Wrote an offline-first, physics-informed ML runtime (Beer-Lambert, Ridge Regression) that measures absorption from 400–700 nm and reads out phosphate and lead levels
- Checks results against BIS IS 10500:2012 drinking-water limits, gives voice readouts in Hindi and English, and logs geotagged results

Ghost: Through-Wall Fall Detection on a WiFi Chip | Python, PyTorch, ESP32, React, Three.js | github.com/yashclouded/ghost

- Built an end-to-end pipeline: ESP32 firmware streams WiFi channel-state information into a CNN with 96% test accuracy and 97.8% fall recall, gated by a debounced state machine
- Added an LLM-written caregiver alert system (with a deterministic fallback) and a live 3D digital twin of the sensing pipeline
- Designed a parametric CAD enclosure and circuit for a sensor node costing about ₹430 (~\$5) per room

NextBench: Campus Marketplace | Kotlin, Android | 300+ active users

- Built a verified-student marketplace with buying and selling, stories, clubs and messaging. The web platform is live, and the native Android app launches in October 2026

Codiva: VS Code Extension | TypeScript | 5.0 rating, 83 users

- Gamified coding extension with a distraction-free Pomodoro timer and analytics on coding habits

Alem: Lightweight Notes App | Python | under 25 MB

- Built for low-resource devices, with rich text, AI-powered semantic search, tagging and secure sharing

Awards & Community

- 2nd Place, Startup Ashoka 2026 (LISA)
- Best App Winner, Codeday Lucknow | 1st Place, Phaser Hackathon | 2nd Place and Best Simulated Circuits, STEMist Hackathon
- Lead Organizer, Scrapyard Lucknow: team building, sponsorship outreach and event execution
- Mentor, NASA Space Apps Challenge Lucknow (400+ participants)
- Partnered with India Innovates, a hackathon with 28,000+ registered participants, hosted at Bharat Mandapam
- Chapter Member, STEMist Lucknow: taught computer science to underprivileged students

Skills

Languages: Python, Kotlin, TypeScript/JavaScript

Web & Mobile: React, Next.js, Node.js, Firebase, Prisma, Android (Kotlin)

AI/ML & Embedded: PyTorch, CNNs, LLM integration, semantic search, ESP32, Raspberry Pi, Three.js